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Brief Report

Dissemination of *Vibrio cholerae* O1 isolated from Odisha, India

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Summary

The present study reported the antimicrobial susceptibility trends, virulence genes, and drug resistance genes of *Vibrio cholerae* O1 strains isolated from outbreaks and epidemics over two and half decades (1995–2019) from Odisha, India. Antimicrobial susceptibility testing was performed by disc diffusion method. Virulence and drug resistance genes were detected by multiplex PCR assays. All *V. cholerae* O1 strains were sensitive to gentamicin, chloramphenicol, norfloxacin and ciprofloxacin while resistant to one or more antibiotics used. About 90% of the isolates of *V. cholerae* O1 carried antibiotic drug resistant genes (*SulII*, *dfrA1* and *strB*) and SXT elements and the results correlated with the phenotypic antibiotic data obtained through disc diffusion assay. The *tcpA* Haitian variant *V. cholerae* O1 first appeared in 1999, gradually showing its increasing number upto 2019. TcpA El Tor strains only prevailed from 1995 to 2006; whereas the *tcpA* classical strains of *V.cholerae*O1 were found in less number from 1995 to 2016. Two multiplex PCR assays confirmed the presence of various toxigenic and virulence genes (*toxR*, *ompU*, *ace*, *rtxC*, *ctxA*, *tcpA*, *rfbO1* and *ompW*) in all isolate of *V. cholerae* O1 strains. The present findings demonstrated the origin and spread of Haitian variants *tcpA* in *V. cholerae* O1 strains over two and half decades.

Introduction

Cholera is an acute watery dirrhoeal disease caused by the gram negative bacterium *Vibrio cholerae*. It still continues to be an important cause of human infection and some time cause devasting out breaks globally with high morbidity and mortality (Joo, 1974). To date, more than 200 serogroups of *V. cholerae* are reported, but particularly O1 and O139 caused cholera out breaks and epidemics throughout the world (Kaper *et al*., 1995). The *V. cholerae* O1 serogroup has two biotypes, namely Classical and El Tor. To date, seven different pandemics are believed to have experienced since 1817. Among these, the causative agent of the first six were classical biotype strains, where as the ongoing seventh pandemic has been caused by El Tor biotype (Safa *et al*., 2010). El Tor and its number of variants have been implicated in numerous massive outbreaks worldwide and become prevalent in some developing and poor countries with limited access to safe drinking water and proper sanitization (Chin *et al*., 2011).

The emergence and spread of antimicrobial resistant (AMR) *V. cholerae*, especially those resistant to tetracycline, nalidixic acid and trimethoprim-sulfanethoxazole has been reported since the 1980s (Ghosh and Ramamurthy, 2011). Resistance to these antimicrobials has been strongly associated with the presence of integrative and conjugative elements (ICEs) of the SXT family (Sarkar *et al*., 2019). In recent years multidrug resistant in *V. cholerae* has been reported after carrying a conjugative plasmid by strains (Falbo *et al*., 1999). Other genetic elements like SXT element and class 1 integron have also been reported as carrying the genetic determinants for antimicrobial resistance (Iwanaga *et al*., 2004). The SXT genetic element plays an important role in the acquirement of antibiotic resistance and also important to anticipate for the presence of *dfrA1* (O1-specific Tmp resistance), *strB* (StrB resistance) genes, *sulII* (encoding Sul resistance) and the SXT element in *V. cholerae* strains (Kumar *et al*., 2010).

The major virulence factors in toxigenic *V. cholerae* strains are cholera toxin (CT), which is encoded by a CTXAB gene and the important determinant of pathogenesis is the toxin co-regulated pilus (TCP), which is encoded by a gene TCPA that is responsible for efficient colonization by the bacterium present in the human

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intestinal tract and serves as CTX profile (Waldor *et al*., 1996). The whole genome sequencing of *V. cholerae* O1 strain isolated from the Haiti cholera outbreak revealed a single nucleotide polymorphism (SNP) at nucleotide position 266 (amino acid position 89) at the *tcpA* gene (Chin *et al*., 2011). The analysis of *tcpA* gene in *V. cholerae* O1 El Tor strains isolated from Kolkata revealed that the Haitian (HCT) variant TCPA in *V. cholerae* O1 El Tor strain emerged in 2003 and showed its full dominance from 2004 to 2012 (Ghosh *et al*., 2014).

In recent years, in spite of having significant advances, it is difficult to understand the pathogenesis of diarrhoeal diseases caused by *V. cholerae*, endemic cholera still causal organism of many people death in Odisha and West Bengal, India. Vibrio species undergo different genetic assortment and reassortment for their better adaptability, which account for antibiotic resistance. Sometimes, the Vibrio species were also shown resistant to more than one antibiotic. Multidrug resistant *V. cholerae* strains were first time reported in Rayagada district of Odisha state in 2007 (Pal *et al*., 2010). Tetracycline-resistant *V. cholerae* O1 strains were reported during 2010 in the same place from Odisha which became sensitive during 2007 in the same area (Kar *et al*., 2015). Fluroquinolone-resistant *V. cholerae* strains were reported in South India in 2002 (Krishna *et al*., 2006). Different recent studies reported limited phenotypic, genotypic and virulence characteristics of *V. cholerae* O1 strains associated with different outbreaks of cholera in Odisha, India (Kar *et al*., 2015; Pal *et al*., 2017). However, no studies have been reported to document the virulence and antibiotic resistance genes particularly the *tcpA* gene of Haitian variants of *V. cholerae* O1 isolated from this region. Therefore, the present study was envisaged to understand the antibiotic susceptibility patterns, virulence factors, drug resistance, emergence and dissemination of Haitian variant *tcpA* gene in *V. cholerae* O1 strains associated with endemic and out breaks of cholera in the eastern part of India. Representative *V. cholerae* strains were isolated for two and half decades (1995–2019) from diarrhoea patients and environmental sources in Odisha and these isolates were subjected to different microbiological and molecular analysis.

Results

A total of 1380 *V. cholerae* O1 strains included in this study were isolated during 1995–2019 from different diarrhoea patients and environmental samples. All the *V. cholerae* strains included in this study produced colonies typical of *V. cholerae* on TCBS agar media. All strains were reacted positively with *V. cholerae* O1 monoclonal antibody but not with the O139 antiserum.

Analysis of antibiogram profiles and antibiotic resistance genes

All *V. cholerae* O1 strains subjected to 11 different antibiotics revealed multidrug resistant (MDR) strains showing resistance to five to six antibiotics. All the tested 1380 *V. cholerae* O1 strains showed resistances to different drugs throughout the study period were streptomycin, nalidixic acid, ampicillin, co-trimoxazole, and furazolidone; while resistance to tetracycline was lowest. All the tested *V. cholerae* O1 strains were susceptible to gentamicin, chloramphenicol, norfloxacin, and ciprofloxacin (Fig. 1).

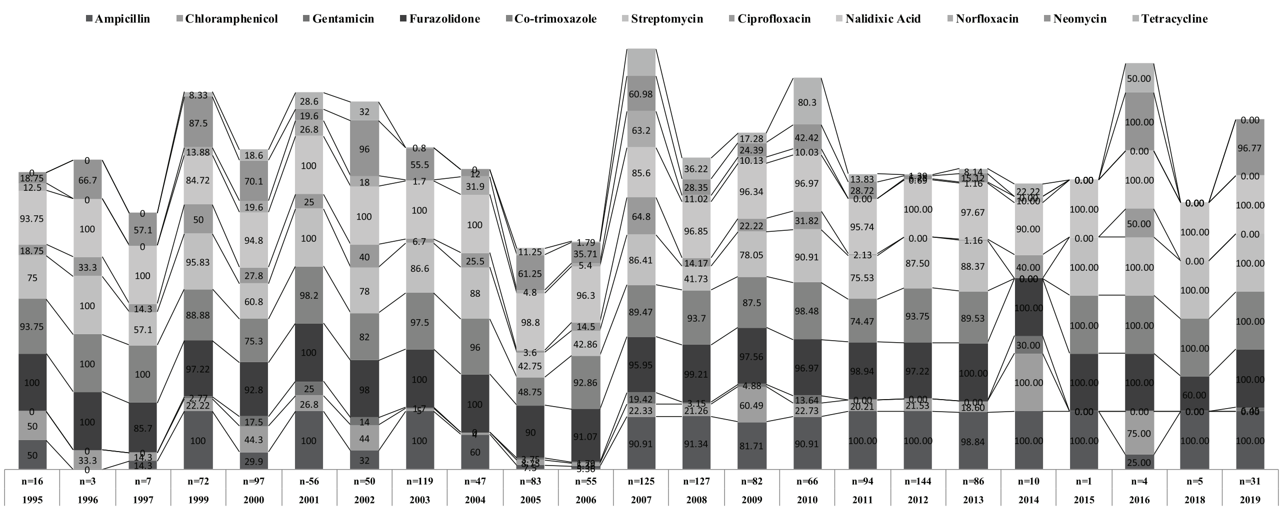

Fig. 1. Comparative resistance profiles of *V. cholerae* O1 against different antibiotics. Numbers in the bar represent percentage to the particular antibiotics.

Some selected *V. cholerae* O1 strains from laboratory stocks were examined for the presence of different antibiotic resistance genes and SXT element. Forty strains of *V. cholerae* O1, 5 each from 2007 and 2008, 2 in 2009, 3 in 2011, 8 in 2012, and 15 in 2013, 1each from 2014 and 2016 gave negative PCR results for *sulII*, *dfrA*1, *strB* genes and SXT element. While testing the antibiotic susceptibility test, these were sensitive to all antibiotics, whereas 142 strains of *V. cholerae* O1, 9 were isolated in 2000, 13 in 2003, 1in 2006, 33 in 2007, 15 in 2008, 21 in 2009, 8 in 2010, 17 in 2011, 9 in 2012, 5 in 2013, 8 in 2014, 1 in 2015, 2 in 2016 produced positive results by PCR assay and amplified a 626 bp fragment of *SulII*, a 278 bp fragment of *dfrA1*, a 515 bp fragment of *strB* and a1035 bp fragment of the SXT element. All of these strains showed resistance to ampicillin, furazolidone, nalidixic acid, streptomycin, trimethoprime and trimethoprime-sulphamethoxazole. One hundred eighty nine strains of *V. cholerae* O1, 6 in 1995, 4 in 1997, 1 in 1998, 38 from 2000, 23 in 2003, 21 in 2005, 10 in 2006, 34 in 2007, 15 in 2008, 21in 2009, 8 in 2010, 8 in 2019 showed positive results by PCR assay and amplified a 278 bp fragment of *dfrA1*, a 515 bp fragment of *strB* and a 1035 bp fragment of the SXT element. All of these *V. cholerae* O1 strains showed resistance to ampicillin, furazolidone, nalidixic acid, streptomycin, and trimethoprim. Thirteen strains of *V. cholerae* O1, seven from 2003 and six from 2010 gave positive results by PCR assay which amplified a 626 bp fragment of *SulII*, a 278 bp fragment of *dfrA1* and a1035 bp fragment of the SXT element. All of these *V. cholerae* O1 strains showed resistance to ampicillin, furazolidone, nalidixic acid, trimethoprim and trimethoprime –sulphamethoxazole. Fifteen strains of *V. cholerae* O1, six in 2006, seven in 2009 and one each from 2007and 2012 gave positive results by PCR assay, which amplified a 278 bp fragment of *dfrA1* and a 515 bp fragment of *strB*. These strains showed resistance to streptomycin and trimethoprim (Table 1).

Analysis of virulence genes

A total 306 *V. cholerae* O1 strains randomly selected from different years were examined for different toxigenic and pathogenic genes by two separate mPCR assays. All strains carried *rfbO1* gene for confirmation of serogroup O1. Most of the strains carried virulence genes for *V. cholerae* O1 with 96% (294/306) being positive for subunitA of cholera toxin gene (*ctxA*), 91% (279/306) positive for *ompU*, 94% (288/306) positive for *toxR*, 85% (260/306) positive for *ace*, 88% (270/306) positive for *ompW*, 89% (274/306) positive for *rtxC* and 89% (274/306) positive for toxin co-regulated pilus gene (*tcpA*) (Table 2).

Analysis of tcpA gene

The *tcp*A gene-specific PCR assay was carried out in all *V. cholerae* O1. The PCR assay yielded a 167 bp amplicon with their respective separate forward primers. During this PCR assay, classical strains did not yield any amplicon in any of the PCR assays due to significant difference in the classical *tcpA* from El Tor *tcpA*. TCPA El Tor showed dominance from 1995 to 2006 after which it completely disappeared. The first appearance of Haitian *tcp*A *V. cholerae* O1 noticed in 1999 and gradually increased upto 2019. Similarly classical *tcpA s*howed its dominance along with the Haitian *tcp*A in different years like 2011 to 2013 and 2016 only. Both Haitian and Classical *tcpA* of *V. cholerae* O1 completely displaced El Tor *tcpA* from 2007 to 2019 (Fig. 2). For further confirmation of PCR-based results for *tcpA* gene, 21 representative strains that resulted positive amplicons for El Tor, Classical and Haitian *tcpA* genes were selected for DNA sequencing of whole *tcpA* gene. The amino acid sequences of all the sequenced strains were identical to the deduced amino acid sequence of the whole *tcpA* of the El Tor reference strain (N16961) except for asparagine to serine (Asn→Ser)substitution at the 89th position of whole *tcp*A (or 64th amino acid of mature *tcpA*) the sequence encompassing the single peptide (Fig. 3).

Table 1. Multiplex PCR assay showing antibiotic resistance genes of *V. cholerae* O1 strains from 1995 to 2019.

Sl. No	No. of strains	Year of isolation (no. of clinical strains) [no.of environmental strains]	Resistance profiles	Strains showing presence of genes encoding			
				*str*B	*SulII*	*dfrA1*	SXT
1	40	2007(5), 2008 (5), 2009 (2), 2012 (8), 2013 (15), 2014 (1), 2016 (1)	Sensitive to all antibiotics	–	–	–	–
2	142	2000 (9), 2003 (13), 2006 (1), 2007 (30) [3], 2008 (15), 2009 (15) [6], 2010 (4) [4], 2011(16)[1], 2012 (8)[1], 2013 (5), 2014 (8), 2015 (1), 2016 (1)[1], 2019[2]	Am, Fr, Na, Str, Tmp, Tmp-Sul	+	+	+	+
3	190	1995 (6), 1997 (4), 1998 (1), 2000 (38), 2003 (23), 2005 (21), 2006 (10), 2007 (33)[1], 2008 (15), 2009 (21), 2010 (9), 2019 (8)	Am, Fr, Na, Str, Tmp	+	–	+	+
4	13	2003 (7), 2010 (6), 2007 [1]	Am, Fr, Na Tmp, Tmp-sul	–	+	+	+
5	15	2006 (6), 2009 (6) [1], 2007 (1), 2012 (1)	Str, Tmp	+	–	+	–

Table 2. Virulence and toxigenic genes of *V. cholerae* O1 strains isolated from 1995 to 2019.

		Source	mPCR I				mPCR II			
Year of isolation	Number of strains	Clinical Environmental	*ompW*	*ctxA*	*rfb*O1	*tcpA*	*ompU*	*rtx*C	*tox*R	*ace*
1995	6	6	5(83%)	4(66%)	6(100%)	4(66%)	4(66%)	4(66%)	4(66%)	3(50%)
		0								
1996	2	2	2(100%)	2(100%)	2(100%)	2(100%)	2(100%)	2(100%)	2(100%)	2(100%)
		0								
1997	4	4	4(100%)	3 (75%)	4(100%)	3 (75%)	3 (75%)	3 (75%)	3 (75%)	2 (50%)
		0								
1999	7	7	6(85%)	5(71%)	7(100%)	5(71%)	5(71%)	5(71%)	5(71%)	4(57%)
		0								
2000	24	24	23(95%)	24(100%)	24(100%)	22(91%)	24(100%)	21(87%)	23(95%)	21(87%)
		0								
2001	20	20	17(85%)	19(95%)	20 (100%)	18(90%)	19(95%)	17(85%)	19(95%)	17(85%)
		0								
2002	25	25	23(92%)	23(92%)	25(100%)	24(96%)	23(92%)	22(88%)	23(92%)	22(88%)
		0								
2003	22	22	19(86%)	21(95%)	22(100%)	19(86%)	21(95%)	20(90%)	21(95%)	20(90%)
		0								
2004	5	5	4(80%)	4(80%)	5 (100%)	3(60%)	4(80%)	3(60%)	5(100%)	4(80%)
		0								
2005	24	24	20(83%)	23(95%)	24(100%)	21(87%)	23(95%)	22(91%)	22(91%)	23(95%)
		0								
2006	7	7	5(71%)	7(100%)	7(100%)	5(71%)	7(100%)	7(100%)	7(100%)	6(85%)
		0								
2007	20	15	15(100%)	15(100%)	15(100%)	15(100%)	15(100%)	15(100%)	15(100%)	14(93%)
		5	5(100%)	5(100%)	5(100%)	5(100%)	5(100%)	5(100%)	5(100%)	4(80%)
2008	19	19	19(100%)	19(100%)	19(100%)	19(100%)	19(100%)	16(84%)	19(100%)	15(78%)
		0								
2009	19	12	12(100%)	12(100%)	12(100%)	12(100%)	12(100%)	12(100%)	12(100%)	10(83%)
		7	7(100%)	7(100%)	7(100%)	7(100%)	7(100%)	7(100%)	7(100%)	6(85%)
2010	13	9	8(88%)	9(100%)	9(100%)	9(100%)	7(77%)	9(84%)	9(92%)	7(77%)
		4	3(75%)	4(100%)	4(100%)	4(100%)	2(50%)	4(84%)	4(92%)	3(75%)
2011	20	14	12(85%)	14(100%)	14(100%)	13(92%)	10(71%)	11(78%)	12(92%)	11(78%)
		6	4(66%)	6(100%)	6(100%)	4(66%)	3(50%)	4(66%)	5(83%)	4(66%)
2012	20	13	12(92%)	13(100%)	13(100%)	12(92%)	11(84%)	13(100%)	13(100%)	11(84%)
		7	5(71%)	7(100%)	7(100%)	6(85%)	5(71%)	7(100%)	6(85%)	5(71%)
2013	19	19	10(52%)	19(100%)	19(100%)	12(63%)	19(100%)	16(84%)	18(94%)	17(89%)
		0								
2014	11	11	11(100%)	10(90%)	11(100%)	11(100%)	10(90%)	10(90%)	10(90%)	10(90%)
		0								
2015	1	1	1(100%)	1(100%)	1(100%)	1(100%)	1(100%)	1(100%)	1(100%)	1(100%)
		0								
2016	3	2	2(100%)	2(100%)	2(100%)	2(66%)	2(100%)	2(100%)	2(100%)	2(100%)
		1	1(100%)	1(100%)	1(100%)	1(100%)	1(100%)	1(100%)	1(100%)	1(100%)
2018	3	3	3(100%)	3(100%)	3(100%)	3(100%)	3(100%)	3(100%)	3(100%)	3(100%)
		0								
2019	12	10	10(100%)	10(100%)	10(100%)	10(100%)	10(100%)	10(100%)	10(100%)	10(100%)
		2	2(100%)	2(100%)	2(100%)	2(100%)	2(100%)	2(100%)	2(100%)	2(100%)

Thus, the DNA sequencing results of the *tcpA* gene confirmed our PCR results. The whole *tcpA* gene sequences of El Tor, Haitian and Classical *V. cholerae* O1 strains were deposited in GenBank under the accession no. MT675530-MT675534, MT670106-MT670110 and MT663512-MT663521.

Discussion

From this study, it was confirmed that all the clinical and environmental *V. cholerae* isolates from different cholera outbreaks reported in Odisha between 1995 and –2019 belonged to serogroups O1.Year-wise phenotypic results revealed that *V. cholerae* O1 isolated between 1995 and 2019 showed dominance of Ogawa serotype over Inaba (Supplementary Table S1). Similar results like dominance of Ogawa also found among the *V. cholerae* isolates in Ghana (Eibach *et al*., 2016). The effective use of antibiotics reduces upto 50% duration of diarrhoea and volume of stool loss and shedding of variable organisms in stool by patients within 1–2 days (Folster *et al*., 2014). Eleven antibiotics recommended from previous studies (ampicillin, chloramphenicol, ciprofloxacin, co-trimoxazole, furazolidone, gentamicin, neomycin, norfloxacin, nalidixic acid, streptomycin, and tetracycline) were used in this study and revealed a loss of sensitivity to many of them including nalidixic acid, streptomycin, co-trimoxazole, furazolidone, ampicillin and neomycin, which were used as the first line of treatment for cholera. Similar findings have been reported in Ghana (Abana *et al*., 2019) and Democratic Republic of Congo (Miwanda *et al*., 2015). The highest percentages of sensitivity of these isolates were recorded in this study against gentamicin, chloramphenicol, norfloxacin, tetracycline and ciprofloxacin. Similar trends have also been reported from other studies in Ghana (Abana *et al*., 2019) and Nepal (Shrestha *et al*., 2015). Similarly nalidixic acid showed 100% resistance to both clinical and environmental isolates in Kathamandu city in Nepal (Greenland *et al*., 2016) and South America (Abana et al., 2019). Resistance patterns of antibiotics in *V. cholerae* are known due to rapidly fluctuation in nature, which unstably carry plasmids that confer resistance and the absence of selective pressure from antibiotics which naturally resides in the organisms in the aquatic environment to the organisms. Our study also showed dramatic fluctuations in resistances to routinely used antibiotics. Nalidixic acid, co-trimoxazole and flurazolidone showed higher percentage of resistance (90%–100%) since 1995 (Fig. 1). The similar types of results of high percentage of resistances to these antibiotics were also reported from Nepal (Rijal *et al*., 2019), Zambia (Mwansa *et al*., 2007), Mozambique (Dengo-Baloi *et al*., 2017) and Haiti (joltund-Karlsson *et al*., 2011). Tetracycline-resistant strains of *V. cholerae* O1 were reported in 2010 from tribal areas of Odisha after which reversal to this drug was observed (Pal *et al*., 2018). The extensive use of these antibiotics for the treatment of other infectious diseases in Odisha other than cholera could be the main reason for resistance to these antibiotics. It could be also due to the acquisition of SXT elements. SXT elements having important features that carry multidrug-resistant genes transfer between bacteria (Wang *et al*., 2016). All *V. cholerae* O1 isolates were resistant to one or more antibiotics as observed in the study which indicated the emergence of multidrug resistance (MDR) strains. Similar types of MDR strains of *V. cholerae* O1 in epidemic period have also been reported from Nepal (Pun *et al*., 2013), Bangladesh (Faruque *et al*., 2006), Pakistan (Siddique *et al*., 1991), Ghana (Abana *et al*., 2019) and India (Pal *et al*., 2018). This could be due to either a spontaneous mutation or the horizontal transfer of resistance genes between co-existance of gut coliform or other microflora and *Vibrio* spp. (Wang *et al*., 2016). This might be due to chromosomal DNA changes also (Baranwal *et al*., 2002).

Cholera pathogenesis in *V. cholerae* is a complex process that involves the synergetic action of several genes. All the genes were found together and represented the genome of the filamentous bacteriophage CTXϕ. By using two sets of mPCR, all the isolates between 1995 and 2019 were screened for the presence of various genes involved in the pathogenicity and toxigenicity. The first mPCR I confirmed the presence of *ompW*, *ctxA*, *tcpA* and *rfbO1* genes, while the second mPCR II analysis revealed that all the isolates were positive for *ompU*, *rtxC*, *toxR* and *ace* genes. Cholera toxin (CT) is considered as the most important marker among various toxins produced by *V. cholerae* (Kaper *et al*., 1995). The presence of *rfbO1*gene confirmed the serogroup O1 and toxigenicity of the isolates respectively. The presence of species-specific *ompW* gene confirmed that the isolates were *V. cholerae*. Gram negative bacteria are widely disseminated from others by the presence of RTX toxins, which represent a family of important virulence factors (Coote, 1992). The presence of RTX toxin in *V. cholerae* encodes an acyltransferase (*rtxC*), the presumptive cytotoxin (*rtxA*) and an associated ATP-binding cassette transporter system which is physically linked to the core element in the *V. cholerae* genome (Lin *et al*., 1999). In this study, various other genes like ace and *toxR* were also screened and all the strains were positive for these genes, suggesting the presence of core toxin region in all the isolates of *V. cholerae* O1.

The toxin co-regulated pilus (TCP) present in *V. cholerae* O1 is an essential colonization factor which serves as a receptor for CTXϕ. It is also a homopolymer of the major pilus protein TCP pillin which encoded by *tcpA*. The Classical and El Tor biotype strains of *V. cholerae* O1 differs slightly on the basis of C-terminal domain of the *tcpA* gene. In the present study, some selected isolates of *V. cholerae* O1 were analysed on the basis of DNA sequencing of *tcpA* gene. Some strains differed from the *V. cholerae* O1 El Tor reference strain N16961 due to single mutation at the amino acid position 64 (Aspargine→Serine). Similarly, a change of amino acid at position 64 of TCPA was already reported in the *V. cholerae* serogroup O1 biotype El Tor strains between 2004 and 2012 isolated in Kolkata (Ghosh *et al*., 2014). On the basis of this result, researchers from Kolkata claimed that Haitian variants of *V. cholerae* O1 strains originated in 2003, but our results showed that Haitian variants of *V. cholerae* O1 emerged after the super cyclone of 1999 from Odisha (Pal *et al*., 2017). Finally this findings not only showed a cryptic change or genetic switching of the *tcpA* gene in *V. cholerae* O1 in Odisha but also raises questions about the origin and spread of this variant of *V. cholerae* O1 El Tor. From the present findings, it is clearly evident that the *tcpA* El Tor

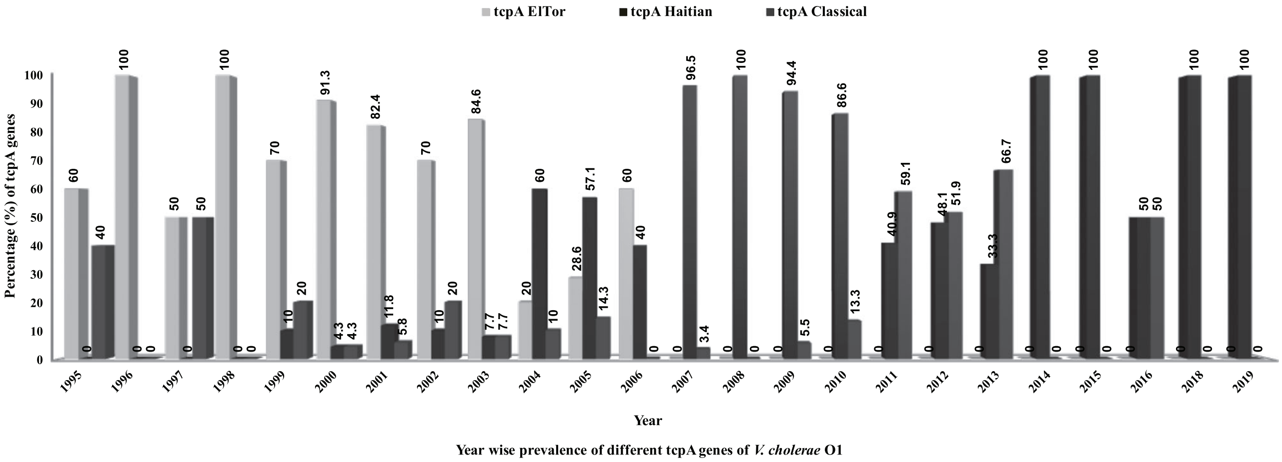

Fig. 2. Year wise prevalence of different types of *tcpA* genes in *V. cholerae* O1 strains (1995–2019).

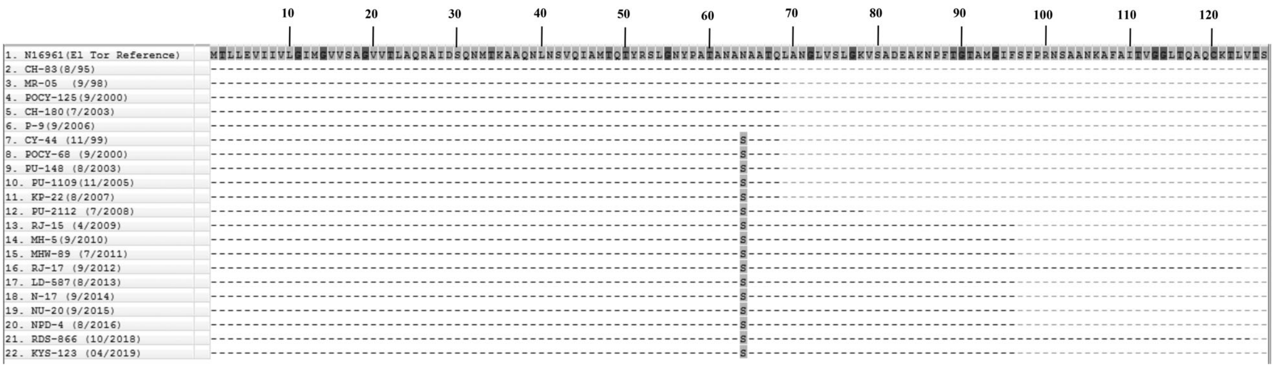

Fig. 3. CLUSTALW alignment of amino acids encoded by the *tcpA* gene of the test *V. cholerae* O1 strains with the sequence from the El Tor reference strain N16961. Dots indicate identical amino acids. The amino acid sequences of *tcpA* of *Vibrio cholerae* O1 strains isolated in Odisha in 1995, 1998, 2000, 2003 and 2006 years showed 100% identity with those of *tcpA* reference strain N16961.

V. cholerae O1 strain prevailed from 1995 to 2006. Whereas *tcpA* classical strain first appeared in 1995, 1997, 1999 and least number in 2000 to 2010; but they showed increasing number from 2011 to 2016 except 2006, 2008, 2014 and 2015. The Haitian variant *tcpA* of *V. cholerae* O1 first emerged in 1999 and gradually increased in number and showed highest percentage in 2007 to 2010, 2014, 2015, 2018 and 2019 (Fig. 1).

Treatment of antibiotic therapy shortens the duration of cholera, but the resistance in *V. cholerae* has also been contributed by the use of antibiotics (Rashed *et al*., 2012). Multidrug-resistant antibiotics are rising common among the *V. cholerae* O1 strains, which mostly associated with the acquisition of different genes or the modification in the antibiotic target genes. From this study, *V. cholerae* O1strains isolated between1995 and 2019 were resistant to streptomycin, furazolidone and chloramphenicol. These antibiotic resistances in *V. cholerae*O1 are frequently associated with SXT element in *V. cholerae* derived ICE element. This transmissible, conjugative and integrating SXT element encoded resistance to sulfamethoxazole, trimethoprim, chloramphenicol and streptomycin (Chomvarin *et al*., 2013). Since 1994, the SXT element was found in *V. cholerae* O1 strains isolated from India, Bangladesh, Mozambique and Laos (Ramachandran *et al*., 2007). In this study, we performed a multiplex PCR targeting integrating and conjugating element (ICE) associated genes *SulII*, *dfrA1*, *StrB* and SXT element, associated with sulfamethoxazole, trimethoprim, streptomycin, nalidixic acid and neomycin. Resistance genes for trimethoprim and sulfamethoxazole were detected in more than 90% of the strains in different years and were absent in less number of strains. Among the resistance genes *dfrA1* for trimethoprim and *sulII* for sulfamethoxazole were detected most frequently in most of the isolates in different years. The genes *SulII* encoding sulfamethoxazole resistance, *dfrA1* encoding dihydropterite synthase and *StrAB*, encoding streptomycin resistance (Sm) were all present in SXT elements were analysed also in this study. Resistance genes to trimethoprim and sulfamethoxazole are often carried by the SXT element and are co-related (Wang *et al*., 2016; Pande *et al*., 2012). All these genes identified in almost all strains were explained and matched the resistance profiles. These results suggested the vast spread of SXT elements in all most all the clinical isolates conferring resistances to these antibiotics. The most important character of SXT element is that it carried multidrug resistance genes. Most of the ICE antibiotic resistance genes are embedded in a composite transposm like element that interrupts the SXT encoded rumAB operon. The antibiotic resistance genes appeared through repeated transposition. Similar results of SXT element were found in most clinical and environmental isolates of *V. cholerae* O1 both from Asian and African countries (Wang *et al*., 2016).

From the present study, it is concluded that resistant to streptomycin, co-trimoxazole, ampicillin, nalidixic acid and furazolidone were higher and stable throughout different years exhibited by *V. cholerae* O1 strains. *V. cholerae* O1 strains possessed the SXT element in different years which suggested the role in the acquisition of antibiotic resistant genes encoding sulfamethoxazole, trimethoprim or streptomycin and showed the wide and rapid spread nature of this element offering the bacterium to be more toxigenic. Two multiplex PCR assays used in this study effectively detected multiple antibiotic and virulence genes which could be also used in the spread of the antibiotic resistance in *V. cholerae* O1in epidemiological and environmental studies. On the basis of *tcpA* gene analysis, it was concluded that the spread of Haitian variant of *V. cholerae* O1 was dominant over classical and El Tor strain. The spread of Haitian variant *tcpA* gene in El Tor *V. cholerae* O1 was observed from 1999 and onwards which needs close monitoring in this region.

Experimental procedures

Serological characterization

A total of 1380 strains of *V. cholerae* O1 isolated from 1995 to 2019 were included in this study. All the strains were obtained from laboratory stocks, which were collected from the diarrhoea patients and environmental water samples from different districts of Odisha during diarrhoea surveillance studies and from different cholera outbreaks and epidemics. Before the initiation of work all *V. cholerae* strains were revived and serologically confirmed with the specific *V. cholerae* O1 antisera (Ogawa and Inaba) (Difco, USA).

Antibiotic susceptibility test

All the *V. cholerae* O1 strains were tested for antimicrobial susceptibility following the method of Bauer (1966) and the clinical and laboratory standards institute (CLSI) (Wayne 2010) guidelines using the antibiotic discs from Difco, USA. Eleven antibiotics were employed like ampicillin (Am: 10 μg), chloramphenicol (Cm: 30 μg), ciprofloxacin (Cf: 5 μg), co-trimoxazole (Co: 25 μg), furazolidone (Fu: 100 μg), gentamicin (Ge: 10 μg), neomycin (N: 30 μg), norfloxacin (Nx:10 μg), nalidixic acid (Na: 30 μg), streptomycin (Str: 10 μg) and tetracycline (T: 30 μg). The resistance profiles of the isolates were determined by measuring the zone of inhibition and comparing it within an interpretative chart to determine the sensitivity of the antibiotics as per the specific chart supplied by the farm.

TCPA gene analysis

The polymerase chain reaction (PCR) assay was carried out broadly on *tcpA* gene to discriminate *V. cholerae* O1strains having Haitian, Classical and El Tor alleles of *tcpA*. The *tcpA*-based PCR assay was used to screen some selected *V. cholerae* O1 clinical and environmental strains isolated from 1995 to 2019 from different districts of Odisha. In this PCR assay, three different primers were used, which included one common reverse primer for both Haitian and El Tor type (tcpA alleles EL-Rev[5'-CCGACTGTAATTGCGAATGC]) and two forward primers (tcpA-F1[5'-CCAGCTACCGCAAACGCAGA-3′] and tcpA-F2[5'-CCAGCTACCGCAAACGCAGG –3′] specific for Haitian and El Tor type *tcpA* alleles respectively(Ghosh *et al*., 2014). PCR assay conditions were as follows: initial denaturation at 94°C for 2 min, 25 cycles of denaturation at 94°C for 1 min, annealing at 56°C for 1 min and extension at 72°C for 2 min and incubation at 72°C for 10 min for the final extension. The amplified products were purified by Wizard® SV (Promega) and subjected for direct sequencing using ABI Big Dye termination cycle sequencing ready reaction kit, following the manufacturer's protocol (Applied Biosystems, USA). *V. cholerae* O1 El Tor (N16961) served as the reference strain. The deduced amino acid sequences of the respective genes from all strains were aligned using CLUSTAL W.

Analysis of virulence genes

Selected *V. cholerae* O1 strains were studied for the detection of virulence, serogroups, pathogenicity and toxigenic genes through two separate multiplex PCR (mPCR) assays. The first mPCR I was used to confirm the presence of *V. cholerae* and its toxigenicity, serogroups and using four sets of primers for genes encoding the outer-membrane protein OmpW (*ompW*), cholera toxin CT (*ctxA*), somatic antigen (*rfb*O1) and toxin coregulated pilus (*tcpA*). The second mPCR II was used to confirm the presence of virulence and toxigenic genes encoding the accessory cholera enterotoxin (*ace*), outer membrane protein (*OmpU*), repeat in toxin protein (*rtxC*) and toxin regulator (*toxR*). The primers were selected to produce amplicons of different sizes for clear identification (Kumar *et al*., 2009). DNA extraction from bacterial cells was done using boiling water. PCR assay was carried out in a reaction mixture of 25 μl containing Taq buffer, 200 nmol dNTPs ml^{-1}, 1.5 mmol $MgCl_2$ ml^{-1}, 1 U *Taq* polymerase, various concentrations of primers specific for each gene (7–12 pmol), 100 ng DNA as template and sterile distilled water up to 25 μl. The thermal cycling conditions were as follows: pre-incubation at 94°C for 2 min; 30 cycles of 1 min at 94°C for denaturation, 1 min at 58°C for annealing and 2 min at 72 C for extension; and incubation at 72°C for 10 min for the final extension.

Assay for antibiotic resistance genes

The presence of antibiotic resistance genes for all *V. cholerae* O1 strains encoding for sulphamethoxazole (*sulII*), trimethoprim (*dfrA1*), streptomycin (*strB*) and the SXT genetic element were determined by using multiplex PCR assay (Ramachandran *et al*., 2007). The multiplex PCR assay was carried out by simultaneous addition of all four primer pairs for *sulII*, *dfrA1*, *strB* and SXT in the same reaction mixture. PCR conditions for amplification were maintained as follows: pre-incubation at 94°C for 2 min; 35 cycles of 1 min at 94°C for denaturation, 1 min at 60°C for annealing and 1 min at 72°C for extension; and incubation at 72°C for 10 min for the final extension.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1 Supplementary Information.